

The Dollar Variance Risk Premium: A Tale of Two Investors

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The risk premium on dollar uncertainty

- ▷ The variance risk premium is the price of insurance against crashes (Carr and Wu 2009; Bollerslev, Tauchen, and Zhou 2009)
- ▷ But exchange rates are **bilateral**: one currency's depreciation is another's appreciation
 - A spike in exchange-rate variance need not signal the same “crash” for everyone
 - The same uncertainty can be good for one and bad for the other.
- ▷ Londono and Zhou (2017): the dollar variance risk premium (DVP) predicts a strong dollar.
 - We reinterpret this from a signed investor-specific perspective.

Two sides of viewing the dollar variance

Two investors: opposite signs!

- ▷ **Standard view:** A single global (US) factor.
 - USD appreciates in bad states (high global/US variance)
 - **US investors**, holding dollar carry position require a premium to hold riskier foreign currency
 - higher dollar variance should predict a **dollar depreciation**
- ▷ **The other side of the market:** **non-US investors**
 - ⇒ Dollar assets introduces valuation uncertainty rather than safety
 - ⇒ Net creditor countries hedge against their own currency appreciation, and this demand rises with dollar uncertainty (Liao and Zhang 2025)
 - ⇒ the same uncertainty should predict a **dollar appreciation**

A tale of two investors

Variance pricing is state-contingent and what matters varies across investors

- ▷ **US investors** – dollar appreciation risk (**Down** states)
 - Downside DVP (DVP^D) measures their hedging demand
 - Predicts future foreign-currency appreciation
 - ▷ **Foreign investors** – foreign-currency appreciation risk (**Up** states)
 - Upside DVP (DVP^U) measures their hedging demand
 - Predicts future dollar appreciation
- ⇒ Skewness Risk Premium: $SRP = DVP^U - DVP^D$
- Relative price of the two tails – a strong (negative) predictor of dollar returns

Why dislike local-currency appreciation?

- 1) **Balance sheet**: International investors are structurally short their own currency (Gourinchas and Rey 2007; Jansen, Shin, and von Peter 2024)
→ a domestic appreciation lowers local-currency value of foreign assets (a negative wealth shock)
- 2) **Trade**: Currency appreciation hurts **export competitiveness**. Firms are net exporters.
- 3) **Financial condition**: Appreciation often associated with tighter financial conditions

⇒ *Currency appreciation is a non-diversifiable risk that requires a premium*

A preview of the main findings

- ▷ The aggregate DVP predicts a subsequent USD appreciation (Londono and Zhou 2017).
- ▷ Decomposing into two tails, they predict with **opposite signs**:
 - Downside DVP (US-investor premium) → foreign-currency appreciation
 - Upside DVP (foreign-investor premium) → dollar appreciation

⇒ SRP (Up — Down) negatively predicts dollar returns, in and out of sample
- ▷ The direction **reverses for safe-haven currencies** (JPY, CHF): global risk loads on their upside.
- ▷ We also find that the upside premium is associated with a country's **net foreign asset position**, while the downside premium reflects common global risk.

Related literature

▷ **Currency VRP and predictability**

- Londono and Zhou (2017); Della-Corte, Ramadorai, and Sarno (2016); Della-Corte, Kozhan, and Neuberger (2021); Kjær and Posselt (2025)

↪ We split the DVP into two tails, argue that each tail contains risk premium required by [two different investors](#)

▷ **Signed / skewness and semi-variance risk premia**

- Feunou, Jahan-Parvar, and Okou (2018); Kilic and Shaliastovich (2019); Bollerslev (2022); Li, Sarno, and Zinna (2025)

↪ We bring the tail decomposition to FX, give it an [economic interpretation](#)

▷ **Dollar, safe assets, and external imbalances**

- Lustig, Roussanov, and Verdelhan (2011, 2014); Maggiori (2017); Gourinchas and Rey (2007); Della-Corte, Riddiough, and Sarno (2016); Liao and Zhang (2025)

↪ Link [net foreign asset position](#) with the dollar variance risk premium

Model setup: a two-country economy

- ▷ Two country model: US (home) and foreign country i .
- ▷ Consumption growth loads on US (z_t) and foreign variance factor ($z_{i,t}$):

$$\Delta c_{t+1} = \mu_t - \lambda' \sqrt{z_t} u_{t+1} - \delta' \sqrt{z_{i,t}} u_{i,t+1}$$

$$\Delta c_{i,t+1} = \mu_{i,t} - \delta'_i \sqrt{z_{i,t}} u_{i,t+1} - \lambda'_i \sqrt{z_t} u_{t+1}$$

- ▷ Variance factors (z) follow square-root dynamics; shocks (u) are standardized Chi-square ($\Rightarrow$ **positively skewed**)
- ▷ With CRRA, the implied SDF:

$$m_{t+1} = a_t + \lambda \sqrt{z_t} u_{t+1} + \delta \sqrt{z_{i,t}} u_{i,t+1}$$

Exchange rates and the currency risk premium

$$\Delta q_{i,t+1} = m_{i,t+1} - m_{t+1}, \quad \text{higher } q_i \Rightarrow \text{foreign appreciation}$$

Currency risk premium:

$$\text{Cov}_t(-m_{t+1}, \Delta q_{i,t+1}) = \underbrace{\lambda(\lambda - \lambda_i)}_{>0} z_t + \underbrace{\delta(\delta - \delta_i)}_{<0} z_{i,t}$$

- ▷ **Home-bias assumption:** $\lambda > \lambda_i$ and $\delta_i > \delta$
 - A positive **US** variance shock $\Rightarrow$ **dollar appreciation**
 - A positive **foreign** variance shock $\Rightarrow$ **foreign-currency appreciation**
- ▷ Each investor cares primarily about their **own** currency's appreciation risk (and requires compensation)

Decomposition into semi-variance risk premia

- ▷ For each currency against USD, we decompose the variance risk premium:

$$CVRP_{i,t} = CVRP_{i,t}^U + CVRP_{i,t}^D$$

- ▷ Each tail loads on **one** factor and prices **one** investor's risk:

$$CVRP_{i,t}^U = \delta(\delta_i - \delta)^2 \nu_i z_{i,t} \quad \text{foreign factor/foreign appreciation risk} \rightarrow \text{higher dollar returns}$$

$$CVRP_{i,t}^D = \lambda(\lambda - \lambda_i)^2 \nu z_t \quad \text{US factor/ dollar appreciation risk} \rightarrow \text{lower dollar returns}$$

- ⇒ The currency risk premium loads on the two tails with **opposite signs**:

$$\text{Cov}_t(-m_{t+1}, \Delta q_{i,t+1}) = \underbrace{\beta_{D,i}}_{>0} CVRP_{i,t}^D + \underbrace{\beta_{U,i}}_{<0} CVRP_{i,t}^U$$

- ⇒ Central testable prediction

- ▷ Skewness risk premium

$$CSR_{i,t} = CVRP_{i,t}^U - CVRP_{i,t}^D$$

Model predictions: safe havens and balance sheets

A multi-country extension: Add a global factor $z_{g,t}$

- ▷ **Safe-haven currencies** (high global exposure, e.g. JPY, CHF)
 - ⇒ These currencies appreciate when global risk rises (loads on upside)
 - ⇒ Opposite predictive signs for safe currencies

- ▷ **Net foreign assets** drive the upside premium:

$$CVRP_{i,t}^U = \gamma^3 (\kappa_{l,i}(x_{i,t}) - d_i)^2 d_i \nu_i z_{i,t}, \quad \text{increasing in } x_{i,t}$$

- Creditor countries take local currency appreciation risk more seriously, because they hedge their own appreciation ⇒ higher upside premium

Upside → local appreciation risk, increasing in NFA

Downside → global risk, independent of NFA

Data sources

- ▷ Currency options and daily returns from Bloomberg (Appreciation = Positive returns)
- ▷ Intraday (5-min) data on currencies and the SPX from histdata.com
- ▷ The G10 currencies against the US dollar:
 - AUD, CAD, CHF, EUR, GBP, JPY, NZD (Jan 2008 – Dec 2022)
 - NOK and SEK (Oct 2008 – Dec 2022)
- ▷ Interest rates from Du and Schreger (2016); NFA from the IMF (IIP)
- ▷ Global risk proxies: US sovereign CDS, global uncertainty (Bekaert, Engstrom, and Xu 2022), and the VIX

Measurement: Variance Risk Premium

- ▷ The currency variance risk premium (CVRP) of pair i /USD:

$$CVRP_{i,t} = E_t^Q[QV_{i,t}] - E^P[QV_{i,t}]$$

- Q from option prices (Vanna-Volga + Bakshi-Kapadia-Madan); P from HAR-RV
- Upside / downside from semi-variances (Barndorff-Nielsen et al. 2010)

- ▷ Aggregate (cross-currency average): $DVP = \frac{1}{N} \sum_i CVRP_i$

$$DVP = DVP^U + DVP^D, \quad SRP = DVP^U - DVP^D$$

- ▷ EVRP: the variance risk premium of the S&P 500 Index

Benchmark: the dollar variance risk premium

$$\Delta q_{i,t+1} = b_0 + b_D DVP_t + b_E EVRP_t + b_I (y_{US,t} - y_{i,t}) + FE_i + e_{i,t+1}$$

(Londono & Zhou, 2017)

Panel A: Using the dollar variance risk premium

	1-month		3-month		6-month		12-month	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DVP	0.947 (0.288)	-3.129 (0.883)	-4.771*** (2.916)	-6.253*** (3.437)	-2.133 (1.464)	-3.405** (2.476)	-1.494 (1.336)	-2.395** (2.237)
EVPR		1.005*** (4.792)		0.365*** (5.345)		0.312*** (2.955)		0.219*** (3.044)
$y_{US} - y_i$	1.847** (2.094)	2.302*** (2.647)	1.600* (1.894)	1.765** (2.108)	1.437* (1.727)	1.577* (1.894)	0.764 (1.394)	0.857 (1.578)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	1602	1602	1602	1602	1602	1602	1602	1602
Adj R^2	-0.001	0.015	0.013	0.020	0.017	0.027	0.022	0.033

- ▷ DVP predicts **dollar appreciation**. EVRP predicts dollar depreciation.

Decomposition: DVP^U and DVP^D

$$\Delta q_{i,t+1} = b_0 + b_U DVP_t^U + b_D DVP_t^D + b_E EVRP_t + b_I (y_{US,t} - y_{i,t}) + FE_i + e_{i,t+1}$$

Panel B: Using the dollar semi-variance risk premium

	1-month		3-month		6-month		12-month	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DVP^U	-34.566*** (2.866)	-31.455** (2.589)	-42.100*** (4.999)	-41.303*** (4.950)	-45.385*** (7.048)	-44.821*** (7.026)	-30.262*** (7.062)	-29.864*** (7.033)
DVP^D	15.001** (2.335)	8.673 (1.277)	8.230*** (2.645)	6.609** (2.088)	13.938*** (4.488)	12.785*** (4.296)	9.475*** (5.022)	8.657*** (4.970)
EVPR		0.889*** (4.127)		0.228*** (3.919)		0.161* (1.674)		0.113* (1.722)
$y_{US} - y_i$	2.150*** (2.572)	2.484*** (2.941)	1.854** (2.285)	1.940** (2.394)	1.764** (2.255)	1.823** (2.317)	0.975* (1.903)	1.014** (1.982)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	1602	1602	1602	1602	1602	1602	1602	1602
Adj R^2	0.006	0.019	0.034	0.036	0.081	0.083	0.089	0.092

Expected pattern if each tail maps to a different marginal investor:

- ▷ DVP^U (required by foreign) $< 0 \Rightarrow$ dollar appreciation
- ▷ DVP^D (required by US) $> 0 \Rightarrow$ foreign-currency appreciation.

The Skewness Risk Premium

$$\Delta q_{i,t+1} = b_0 + b_S SRP_t + b_D DVP_t + b_E EVRP_t + b_I (y_{US,t} - y_{i,t}) + FE_i + e_{i,t+1}$$

Panel C: Using the dollar skewness risk premium

	1-month		3-month		6-month		12-month	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
SRP	-20.758*** (2.846)	-16.699** (2.238)	-19.292*** (4.569)	-18.046*** (4.325)	-23.807*** (6.177)	-22.884*** (6.031)	-16.315*** (7.117)	-15.658*** (7.016)
DVP	-5.783* (1.654)	-8.244** (2.275)	-11.026*** (4.917)	-11.781*** (5.115)	-9.867*** (6.638)	-10.426*** (7.034)	-6.817*** (6.311)	-7.215*** (6.733)
EVPR		0.931*** (4.299)		0.286*** (4.702)		0.211** (2.210)		0.149** (2.312)
$y_{US} - y_i$	2.184*** (2.631)	2.539*** (3.048)	1.913** (2.354)	2.022** (2.493)	1.820** (2.314)	1.899** (2.402)	1.016* (1.969)	1.069** (2.078)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	1602	1602	1602	1602	1602	1602	1602	1602
Adj R^2	0.005	0.019	0.033	0.037	0.077	0.081	0.088	0.093

- Controlling for the level of dollar uncertainty, it is **the relative price of its two tails** that predicts dollar returns.

Robustness: the trade-weighted dollar index

- ▷ Time-series regressions on the (negative of) dollar index :

$$-\Delta DXY_{t+1} = b_0 + b_U DVP_t^U + b_D DVP_t^D + b_E EVRP_t + b_I \text{diff}_t + e_{t+1}$$

- ▷ Free of cross-currency correlation concerns.

Panel A. Using the Dollar Semi-variance Risk Premium

	1-month		3-month		6-month		12-month	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
DVP^U	-37.883 (1.428)	-35.940 (1.320)	-35.108** (2.488)	-34.820** (2.467)	-35.900*** (3.545)	-35.583*** (3.520)	-23.724*** (3.250)	-23.482*** (3.217)
DVP^D	20.517* (1.826)	16.721 (1.364)	8.917* (1.747)	8.356 (1.549)	12.518*** (3.083)	11.896*** (3.051)	7.648** (2.582)	7.169*** (2.628)
$EV RP$		0.554** (2.121)		0.082 (0.809)		0.090 (0.464)		0.069 (0.493)
$\overline{y_{US} - y_i}$	2.172 (1.463)	2.435 (1.562)	1.956 (1.481)	1.995 (1.527)	1.921 (1.570)	1.963 (1.588)	1.230 (1.394)	1.260 (1.443)
Obs.	180	180	180	180	180	180	180	180
Adj R ²	0.019	0.026	0.048	0.044	0.114	0.110	0.113	0.110

- ▷ SRP (Paper) negatively predicts the dollar index, increasing with horizon.

Out-of-sample predictability

- ▷ Rolling-window forecasts of the dollar index (in-sample 2008–2012, OOS 2013–2022).

Panel A. Random Walk without Drift

	Benchmark	DVP ^U and DVP ^D	SRP	DVP
RMSE	0.211	0.210**	0.208**	0.213
R_{OOS}^2		0.012	0.026	−0.021
Clark-West t-test		(1.917)	(2.136)	(0.341)
OOS Obs.		120	120	120

Panel B. Random Walk with Drift

	Benchmark	DVP ^U and DVP ^D	SRP	DVP
RMSE	0.211	0.210*	0.208**	0.213
R_{OOS}^2		0.011	0.025	−0.022
Clark-West t-test		(1.609)	(1.929)	(−0.320)
OOS Obs.		120	120	120

Panel C. Uncovered Interest Rate Parity

	Benchmark	DVP ^U and DVP ^D	SRP	DVP
RMSE	0.211	0.210**	0.208**	0.213
R_{OOS}^2		0.006	0.020	−0.027
Clark-West t-test		(1.939)	(2.122)	(0.328)
OOS Obs.		120	120	120

- ▷ **SRP** is the most robust predictor: positive R_{OOS}^2 and significant Clark–West tests across benchmarks.

Safe vs. risky currencies

Safe currency = currencies that appreciated in times of high global uncertainty.

- ▷ **Risky:** downside > 0 , upside < 0 (as in the aggregate).
- ▷ **Safe:** (CHF/JPY Rinaldo & Söderlind 2010) the upside = global-risk premium; no clear prediction of the downside.

Panel A: Using Semi-variance Risk Premium

	Risky Currencies				Safe Currencies			
	1-month	3-month	6-month	12-month	1-month	3-month	6-month	12-month
DVP^U	-19.328* (1.733)	-35.246*** (5.883)	-40.725*** (8.483)	-30.139*** (8.691)	-0.699 (0.090)	8.394* (1.832)	15.173*** (4.649)	15.015*** (5.867)
DVP^D	9.509* (1.934)	3.473 (1.464)	7.918*** (3.703)	5.530*** (4.406)	7.219 (0.636)	2.863 (0.398)	4.630 (0.880)	-0.901 (0.250)
$y_{US} - y_i$	2.327*** (2.900)	2.268*** (2.657)	2.194*** (2.655)	1.255** (2.411)	1.958** (2.431)	2.212** (2.506)	2.307** (2.547)	1.373** (2.400)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	1602	1602	1602	1602	1602	1602	1602	1602
Adj R^2	0.004	0.043	0.112	0.144	-0.002	0.014	0.061	0.101

Time-series driver: Global uncertainty

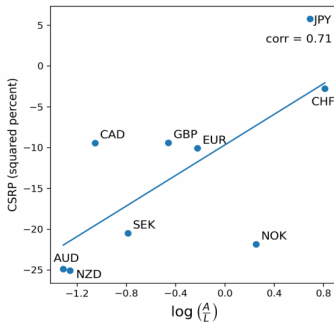
$$SRP_t = \alpha_0 + \alpha_1 GUNC_t + \gamma_c \text{Control}_{t-1} + \epsilon_t$$

Panel A: Skewness Risk Premium

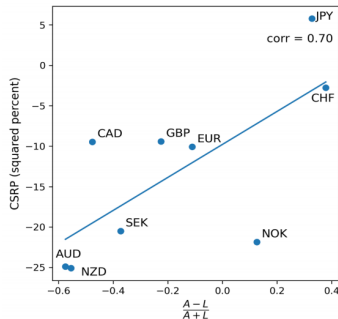
	UNC = BEX				UNC = CDS				UNC = log(VIX)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
UNC_t	-28.606*** (-3.45)	-14.176*** (-2.94)	-12.608** (-2.00)	-11.965* (-1.94)	-0.288*** (-2.67)	-0.403*** (-4.07)	-0.388*** (-3.96)	-0.393*** (-3.84)	-0.894*** (-2.65)	-0.325*** (-2.73)	-0.282* (-1.94)	-0.258* (-1.87)
UNC_{t-3}		-0.660 (-0.17)	-0.102 (-0.03)	0.084 (0.02)		0.145 (1.50)	0.153 (1.51)	0.159 (1.50)		-0.087 (-0.88)	-0.072 (-0.78)	-0.070 (-0.76)
SRP_{t-12}			0.040 (0.47)				0.113 (1.57)				0.052 (0.70)	
DVP_{t-12}^U				0.144 (1.10)				0.071 (0.41)				0.129 (0.97)
DVP_{t-12}^D				-0.044 (-0.55)				-0.114 (-1.64)				-0.056 (-0.81)
Obs.	180	168	168	168	168	156	156	156	180	168	168	168
Adj-R ²	0.120	0.036	0.033	0.032	0.048	0.083	0.090	0.085	0.158	0.030	0.028	0.025

- ▷ SRP falls with all three global-risk proxies – global uncertainty (Bekaert, Engstrom, and Xu 2022), US CDS, and the VIX.
- ▷ In bad times, the cost of downside (dollar) protection surges relative to upside protection.

Cross-section: external imbalance and the upside premium



(a) Using Log Asset-to-Liability Ratio



(b) Using Net Foreign Asset Ratio

Figure 2: External Imbalance and Average Skewness Risk Premium

- ▷ Cross-section of time-series average
- ▷ The average CSRP is correlated with a country's net foreign asset (NFA) position. ($\rho \approx 0.70$).

NFA drives the upside, not the downside

$$CVRP_{i,t}^{U/D} = \alpha + \beta_c NFA_{i,t} + \beta_l (\text{lagged dep.}) + \text{Time FE} + \epsilon_{i,t}$$

Dependent Variable: $CVRP_{i,t}^U$					Dependent Variable: $CVRP_{i,t}^D$				
	$\log(A) - \log(L)$		$(A - L)/(A + L)$			$\log(A) - \log(L)$		$(A - L)/(A + L)$	
	(1)	(2)	(5)	(6)		(1)	(2)	(5)	(6)
$NFA_{i,t}$	0.195*** (3.627)	0.087*** (2.848)	0.092*** (3.689)	0.041*** (2.927)	$NFA_{i,t}$	-0.020 (-0.709)	-0.025 (-0.943)	-0.007 (-0.540)	-0.010 (-0.801)
$CVRP_{i,t-12}^U$		0.254*** (6.649)		0.252*** (6.570)	$CVRP_{i,t-12}^D$		0.026 (0.923)		0.026 (0.928)
Time FE	Yes	Yes	Yes	Yes	Time FE	Yes	Yes	Yes	Yes
Obs.	1602	1494	1602	1494	Obs.	1602	1494	1602	1494
Adj- R^2	0.173	0.485	0.178	0.286	Adj- R^2	0.372	0.280	0.372	0.360

- ▷ The upside rises with NFA (contemporaneous and lagged), no effect on downside
- ▷ Exactly consistent with the [cross-sectional prediction](#) of the model.

Conclusion

- ▶ Pricing currency variance is [a tale of two investors](#): US and foreign investors price different tails.
- ▶ The downside DVP (US-investor premium) predicts foreign-currency appreciation; the upside DVP (foreign-investor premium) predicts dollar appreciation.
- ▶ This is consistent with the downside being the US-investor premium and the upside being the foreign investor premium.
- ▶ The signs reverse for safe-haven currencies.
- ▶ The time-series of downside premium is related to global risk.
- ▶ The cross-section of the upside premium is driven by net foreign asset positions.